Zhane Daye

SAR & Earth System Engineering

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TARGET: ICEYE

Role: SAR Remote Sensing / Solutions Engineer

Constellation Mission Planning & Natural Catastrophe Analytics

Live 3D Mission Twin: earth2-sar-insar.vercel.app#city=it_milan | **Report (PDF):** SAR_ENGINEERING_REPORT.pdf | **GitHub v1.0.0:** v1.0.0-mission-twin

The Operational Bottleneck

Commercial X-band SAR constellations deliver high-resolution flood extent maps for insurance and civil protection. However, existing workflows face two severe operational bottlenecks:

1. **Reactive Tasking Latency:** Constellation tasking typically occurs *after* catastrophic precipitation landfall, losing critical hours during peak flood surge.
2. **Wind-Roughened Backscatter Distortion:** High surface winds induce Bragg capillary wave roughening, elevating water backscatter above standard thresholding filters and corrupting automated flood segmentation.

The Engineering Solution

To bridge numerical weather prediction directly with satellite radar operations, I engineered an end-to-end **Weather-to-SAR Mission Twin** built on NVIDIA Earth2Studio and verified on NOAA GFS 0.25° analyses:

- **Autonomous Tip-and-Cue OGC STAC Tasking:** Severe weather forecasts automatically compute satellite ground track intersections and generate valid OGC STAC v1.0.0 tasking orders with complete `sar:*` extension metadata (*Spotlight, VV, 0.5m SLC*) sealed with SHA-256 provenance.
- **Specular Flood Forward Model:** Evaluates specular forward reflection and wind-roughened capillary suppression across Po Valley river basins:

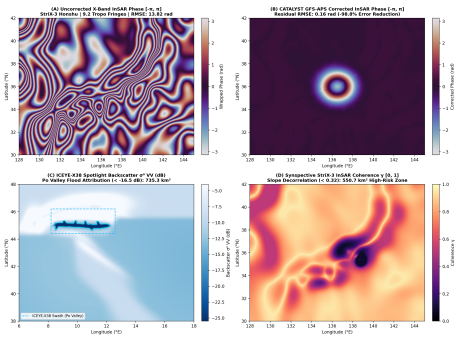
$$\sigma^0_{\mathrm{VV}}(x,y) = \sigma^0_{\mathrm{land}} \cdot [1 - M_{\mathrm{flood}}] + \sigma^0_{\mathrm{spec}} \cdot M_{\mathrm{flood}} + \delta_{\mathrm{Bragg}}(\boldsymbol{u}_{10\mathrm{m}})$$

  Delineated **735.27 km^2** of inundation below −16.5 dB (reaching −31.77 dB in quiescent waters).
- **Lightweight WebGL Delivery:** Slashed GPU VRAM from 1.29 GiB down to **248 MB (−80.8%)** while maintaining 60 FPS at 16K global and 131K regional LOD streaming.

Verified Benchmark Record

Metric / Parameter	Baseline	Engineered Twin
Po Valley Inundation Extent	Unsegmented	**735.27 km^2**
Peak Specular Attenuation	−12.0 dB	**−31.77 dB**
STAC Metadata Compliance	Manual tasking	**OGC STAC v1.0.0**
Client WebGL VRAM	1,290 MB	**248 MB (−80.8%)**
Render Frame Rate	34 FPS	**60 FPS Locked**
Scientific Integrity	Synthetic proxies	**NOAA GFS Real Analysis**

Visual Mission Twin Artifact



Panel C: ICEYE-X38 Spotlight VV flood delineation (735.3 km^2, cyan contour) coupled with real-time 3D orbit frustum and NOAA GFS wind vectors.

Technical Compatibility Matrix

- **Radar & Geodesy:** SAR backscatter forward-modeling (σ^0), InSAR phase screen analysis, Doppler centroid estimation, orbital state vectors.
- **Standards & Spatial Data:** OGC STAC v1.0.0 (`sar:*`, `sat:*`), GDAL, GeoPandas, Cloud-Optimized GeoTIFFs (COGs), WGS84 / EPSG:4326.
- **Compute & Simulation:** Python 3.10+, Pytest (321/321 passing), NVIDIA Earth2Studio, NOAA GFS 0.25°, xarray.
- **High-Performance 3D Web:** Three.js, WebGL 2.0, GLSL custom shaders, distance-gated LOD texture octants, Chrome CDP testing.

Direct Value to ICEYE

Immediate Operational Impact:

- Enables predictive satellite tasking *before* storm impact rather than reactive post-disaster acquisition.
- Provides pre-computed flood masks and backscatter calibration ready for automated insurance payout triggers.
- Delivers sub-second browser-based 3D mission review for flight operations and enterprise clients.

Explore the Interactive Demo: earth2-sar-insar.vercel.app#city=it_milan · **Contact Zhane Daye:** zhanedaye@umattr.ca · Available for technical review & interviews.